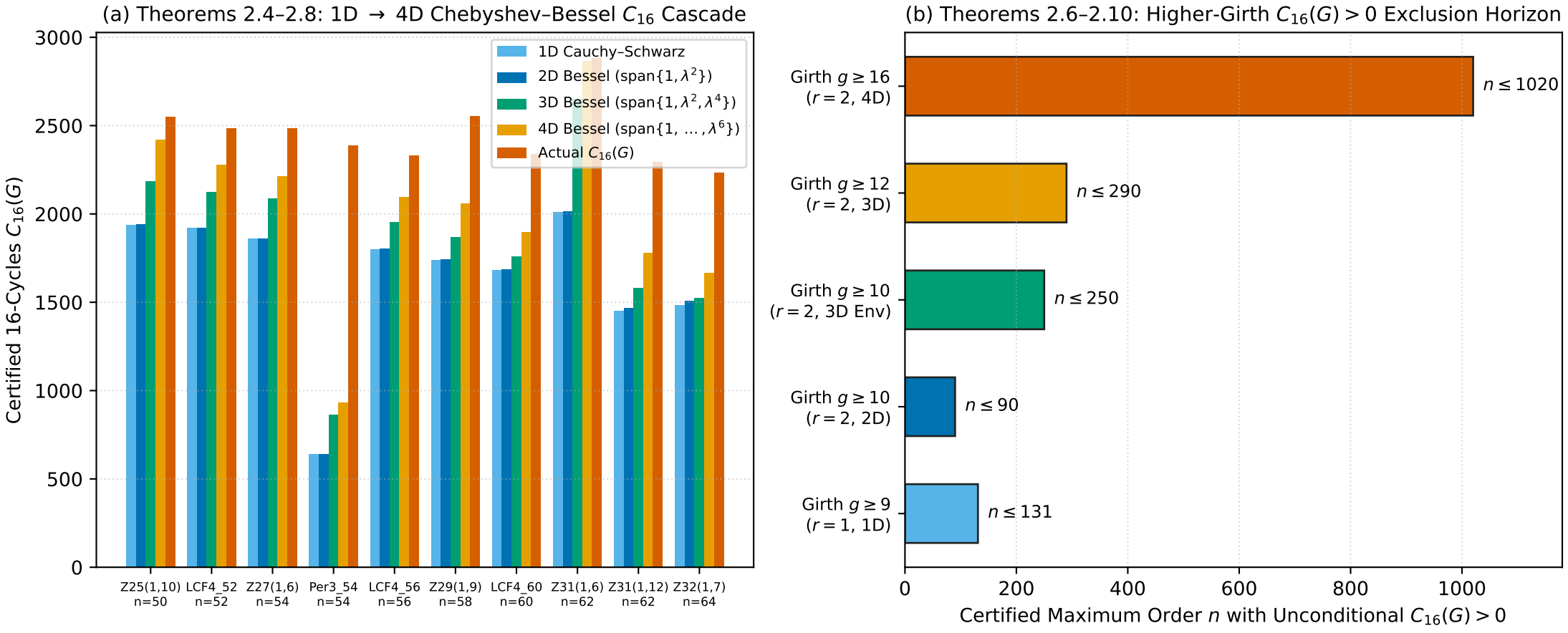
(a) Theorems 2.4–2.8: 1D → 4D Chebyshev–Bessel C_{16} Cascade
Certified 16-Cycles $C_{16}(G)$
1D Cauchy–Schwarz
2D Bessel (span$\{1, \lambda^2\}$)
3D Bessel (span$\{1, \lambda^2, \lambda^4\}$)
4D Bessel (span$\{1, \ldots, \lambda^6\}$)
Actual $C_{16}(G)$
Z25(1,10) n=50
LCF4_52 n=52
Z27(1,6) n=54
Per3_54 n=54
LCF4_56 n=56
Z29(1,9) n=58
LCF4_60 n=60
Z31(1,6) n=62
Z31(1,12) n=62
Z32(1,7) n=64
(b) Theorems 2.6–2.10: Higher-Girth $C_{16}(G) > 0$ Exclusion Horizon
Girth $g \geq 16$ ($r = 2$, 4D)
$n \leq 1020$
Girth $g \geq 12$ ($r = 2$, 3D)
$n \leq 290$
Girth $g \geq 10$ ($r = 2$, 3D Env)
$n \leq 250$
Girth $g \geq 10$ ($r = 2$, 2D)
$n \leq 90$
Girth $g \geq 9$ ($r = 1$, 1D)
$n \leq 131$
Certified Maximum Order n with Unconditional $C_{16}(G) > 0$